why do we use pandas ?

1. data loading
2. data manipulation
3. detecting missing values and filing them

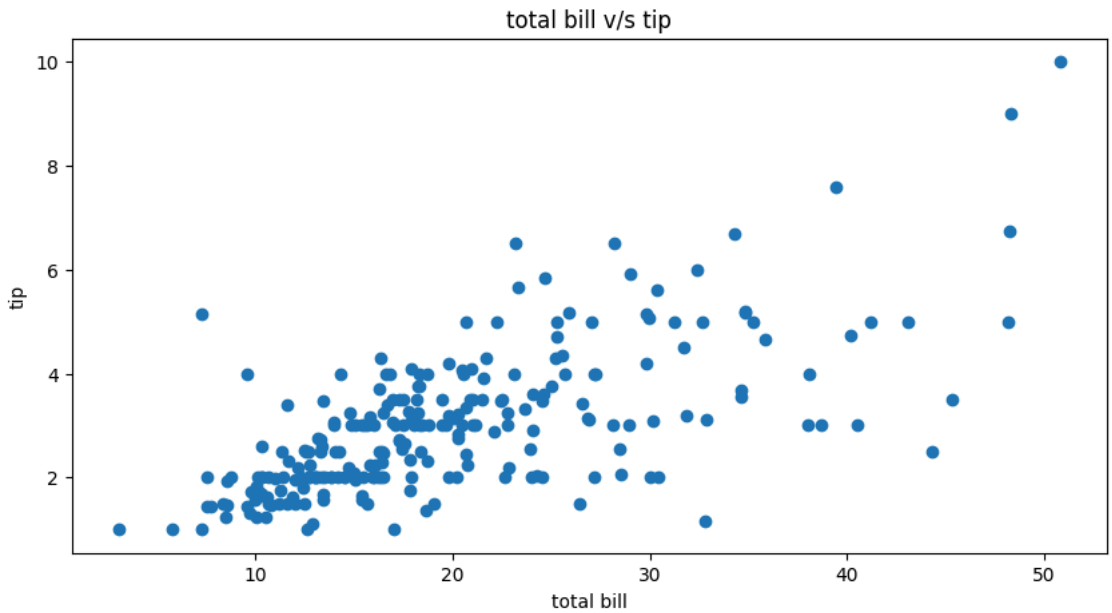
seaborn is used for data visualisation

1. univariate
2. bivariate

matplotlib advanced than seaborn size of figure can be given. plt.figure(figsize=())

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
df=sns.load_dataset('tips')
df

plt.figure(figsize=(10,5))
plt.scatter(df['total_bill'],df['tip'])
plt.title("total bill v/s tip")
plt.xlabel("total bill")
plt.ylabel("tip")
plt.show()   # all these features of making the graph more presentable is all due to matplotlib
```



import pandas as pd df=pd.read_csv("

```
import pandas as pd
df=pd.read_csv("BigBasket Products.csv")
df
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.00	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.00	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.00	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi

inspect

```
df.head()
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.0	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...

```
df.tail()
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
27550	27551	Wottagirl! Perfume Spray - Heaven, Classic	Beauty & Hygiene	Fragrances & Deos	Layerr	199.20	249.0	Perfume	3.9	Layerr brings you Wottagirl Classic fragrant b...
27551	27552	Rosemary	Gourmet & World	Cooking &	Puramate	67.50	75.0	Herbs, Seasonings	4.0	Puramate rosemary is enough to

```
df.shape
```

```
(27555, 10)
```

handle missing

```
df.isnull().sum()
```

	0
index	0
product	1
category	0
sub_category	0
brand	1
sale_price	0
market_price	0
type	0
rating	8626
description	115

dtype: int64

```
df["rating"]=df["rating"].fillna(df["rating"].mean())
df
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
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2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.00	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
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```
df.isnull().sum()
```

	0
index	0
product	1
category	0
sub_category	0
brand	1
sale_price	0
market_price	0
type	0
rating	0
description	115

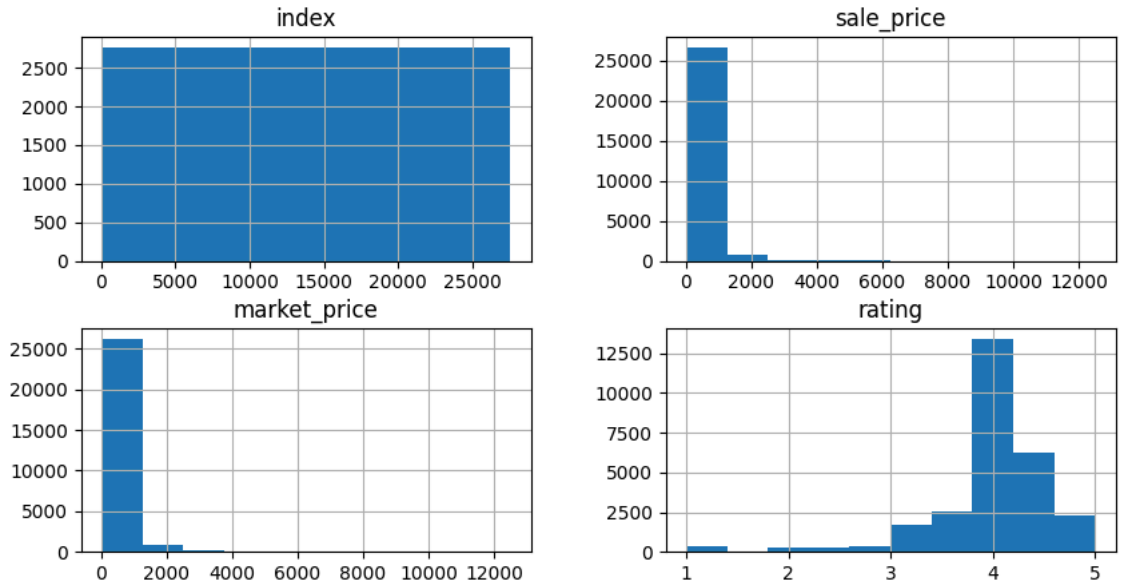
dtype: int64

```
df["product"]=df["product"].fillna("unknown")
df["description"]=df["description"].fillna("unknown")
```

univariate

```
df.hist(figsize=(10,5))
```

```
array([[<Axes: title={'center': 'index'}>,
        <Axes: title={'center': 'sale_price'}>],
       [<Axes: title={'center': 'market_price'}>,
        <Axes: title={'center': 'rating'}>]], dtype=object)
```



bivariate

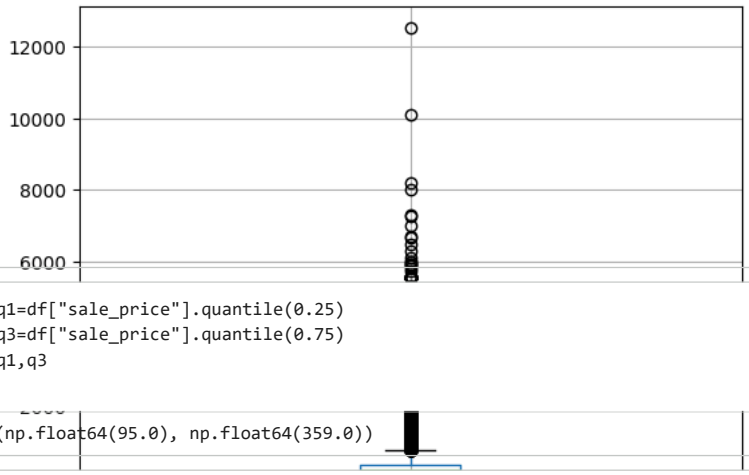
```
import matplotlib.pyplot as plt
plt.scatter(df['sale_price'],df['market_price'])
plt.title("sale_price v/s market_price")

plt.xlabel("sale_price")
plt.ylabel("market_price")
plt.show()
```

outliers

```
df.boxplot(column="sale_price")
```

```
<Axes: >
```




```
q1=df["sale_price"].quantile(0.25)
q3=df["sale_price"].quantile(0.75)
q1,q3
```

```
(np.float64(95.0), np.float64(359.0))
```

```
iqr=q3-q1
iqr
```

```
np.float64(264.0)
```

```
upper=q3+1.5*iqr
lower=q1-1.5*iqr
upper,lower
```

```
(np.float64(755.0), np.float64(-301.0))
```

```
outliers=df[(df["sale_price"]>=upper) | (df["sale_price"]<=lower)]
outliers
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
8	9	Biotin & Collagen Volumizing Hair Shampoo + Bi...	Beauty & Hygiene	Hair Care	StBotanica	1098.00	1098.0	Shampoo & Conditioner	3.50000	An exclusive blend with Vitamin B7 Biotin, Hyd...
47	48	Colour Catcher Sheets	Cleaning & Household	All Purpose Cleaners	Dylon	799.00	799.0	Imported Cleaners	4.00000	1. Prevents Colour Run Accidents Colours that ...
51	52	Peach Syrup	Gourmet & World Food	Drinks & Beverages	Pekers	850.00	850.0	Gourmet Juices & Drinks	4.20000	Pekers peach syrup takes you on a historical t...
69	70	Padded Harness - 3/4 inch, Grey Colour	Kitchen, Garden & Pets	Pet Food & Accessories	Glenand	840.00	840.0	Pet Collars & Leashes	3.94341	These are soft padded harness for your active ...
91	92	Hard Anodised Ezee-Pour Saucepan	Kitchen, Garden &	Cookware & Non Stick	Hawkins Futura	864.50	910.0	Tawa & Sauce Pan	4.60000	Futura Hard Anodised Saucepan

```
df=df[(df["sale_price"]<=upper) & (df["sale_price"]>=lower)]
```

Start coding or generate with AI.

```
df.boxplot(column="sale_price")
```

```
<Axes: >
```